

CE 310 — Week 10 Pre-Lab Quiz

Week: 10 | Topic: Multiple Linear Regression — Concepts and Python/statsmodels Points: 10 questions × 1 pt each = 10 pts Submission: Complete in D2L before class begins. No partial credit for MC.

Instructions

Select the single best answer for each question. The quiz covers multiple linear regression, adjusted R^2 , dummy variables, multicollinearity, interaction terms, and the Python tools you will use in today's lab. Review the Week 10 Background reading before completing this quiz.

Questions

Q1. Simple linear regression uses one predictor to explain variation in an outcome. What does multiple linear regression add over simple linear regression?

- (A) It replaces the least-squares criterion with maximum likelihood estimation
 - (B) It allows two or more predictors to simultaneously explain variation in the outcome, potentially capturing effects that a single predictor cannot
 - (C) It guarantees that $R^2 = 1.0$ if enough predictors are added
 - (D) It removes the need to check whether the relationship between x and y is linear
-

Q2. In Week 9, outdoor temperature explained $R^2 = 0.601$ of the variance in cooling energy — leaving 39.9% unexplained. Which of the following physical factors is the most likely additional predictor that could explain some of the remaining variance?

- (A) The building's street address
 - (B) Whether the building is occupied (business hours vs. nights and weekends)
 - (C) The year the building was constructed
 - (D) The color of the building's exterior walls
-

Q3. Adjusted R^2 (Adj. R^2) differs from R^2 in that it:

- (A) Measures the fraction of variance explained by the model, scaled to the range 0–100%
- (B) Penalizes for adding predictors that do not substantially improve model fit, so it can decrease when a weak predictor is added
- (C) Equals the square of the Pearson r between observed and predicted values
- (D) Is always larger than R^2 because it corrects for sample size bias

Q4. A dummy variable (also called an indicator variable) is:

- (A) A variable with a small standard deviation, indicating it has low predictive power
- (B) A randomly generated variable used to test model stability
- (C) A 0/1 binary variable that encodes membership in a category (e.g., 1 = occupied, 0 = unoccupied)
- (D) A variable that is the square of another predictor, used to model nonlinear effects

Q5. The Model 2 regression equation is: $\text{Cooling_kWh} = -56.87 + 1.026 \times \text{Outdoor_Temp_F} + 17.85 \times \text{BH}$. Without running any code, compute the predicted cooling load for an occupied hour ($\text{BH} = 1$) at $\text{Outdoor_Temp_F} = 90^\circ\text{F}$.

- (A) 53.32 kWh/hr — correct: $-56.87 + (1.026 \times 90) + 17.85$
- (B) 35.47 kWh/hr — this is the unoccupied ($\text{BH}=0$) prediction at the same temperature; forgets to add the BH term
- (C) 110.19 kWh/hr — forgets to add the (negative) intercept
- (D) 17.85 kWh/hr — uses only the BH coefficient, ignoring the intercept and temperature terms entirely

Q6. After fitting the two-predictor model $\text{Cooling_kWh} \sim \text{Outdoor_Temp_F} + \text{BH}$, the regression output shows: $R^2 = 0.697$, BH coefficient = 17.9 kWh/hr, p-value for BH ≈ 0 . Which interpretation is correct?

- (A) R^2 increased from 0.601 to 0.697 primarily because adding any variable always increases R^2 ; the BH coefficient is not meaningful
- (B) Adding occupancy status (BH) increases the fraction of cooling variance explained from 60.1% to 69.7%. Holding outdoor temperature constant, occupied hours require approximately 18 kWh/hr more cooling than unoccupied hours — a result that is highly statistically significant ($p \approx 0$)
- (C) The BH coefficient of 17.9 kWh/hr represents the mean cooling demand during business hours, not an incremental effect
- (D) $R^2 = 0.697$ means the model predicts cooling energy with 69.7% accuracy; the remaining 30.3% of predictions are incorrect

Q7. Multicollinearity in a multiple regression model refers to:

- (A) Having more predictors than observations in the dataset
- (B) A situation where two or more predictors are highly correlated with each other, making it difficult to isolate each predictor's independent effect on the outcome
- (C) Residuals that are not normally distributed
- (D) A model with a very low R^2 value, indicating that none of the predictors explain the outcome

Q8. An interaction term between two predictors x_1 and x_2 is created by multiplying them together ($x_1 \times x_2$). What does a significant interaction term tell you?

- (A) That x_1 and x_2 are collinear and should not both be included in the model
- (B) That the effect of x_1 on the outcome is the same regardless of the value of x_2
- (C) That the effect of x_1 on the outcome changes depending on the value of x_2 — in other words, the slope for x_1 is different at different levels of x_2
- (D) That x_1 and x_2 together explain 100% of the variance in the outcome

Q9. In machine learning, the process of adding new predictors to a model — or creating new variables from existing ones (such as the product of two columns, or a 0/1 dummy variable) — is called:

- (A) Regularization
- (B) Feature engineering
- (C) Cross-validation
- (D) Bootstrapping

Q10. In Python, the expression `smf.ols('Cooling_kWh ~ Outdoor_Temp_F + BH', data=df).fit()` does which of the following?

- (A) Computes the Pearson correlation between `Outdoor_Temp_F` and `Cooling_kWh`, ignoring `BH`
- (B) Fits a multiple linear regression model using `Outdoor_Temp_F` and `BH` as predictors of `Cooling_kWh`, and returns a fitted model object from which coefficients, R^2 , and p-values can be extracted
- (C) Creates a new DataFrame column called `Cooling_kWh` by multiplying `Outdoor_Temp_F` by `BH`
- (D) Fits a logistic regression model to predict whether `BH` equals 0 or 1 from `Cooling_kWh`